Fiscal Year:	FY 2019	**Task Last Updated:**	FY 10/12/2018
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 12/11/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	NOTE: Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and		

	minimize pollutants. This effort is a collaborative effort between the PI (Principal Investigator), the Co-Is (Co-Investigators), and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, and Peter Sunderland directing experiments and numerical simulations at University of Maryland.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	Many laser-induced breakdown spectroscopy (LIBS) tests were performed. The system was optimized and the signal/data processing methodology was improved. We obtained calibration data of the relation between atomic ratio and intensity ratio of major elements (C,H,O,N) from a McKenna flat-flame burner. We reduced the burden on the transition stage by cutting down the distance from the measured point to the transition stage (for ethylene-air counter-flow burner). We measured the C/O, H/O, C/H atomic ratios along the center line of ethylene-air counter-flow diffusion flame. We used the acoustic system (LIBT) to measure the temperature profile on the center line of an ethylene-air counter-flow diffusion flame. All the results have been compared to the OPPDIF simulation and Wengdong's previous data, and we have verified that the LIBS measurement is reliable and repeatable. At Washington University in the last two months, we have performed experiments to show the PM emissions from the pure ethylene flame and comparing PM (particulate matter) data of newly bought low pressure impactor [Electrical Low-Pressure Impactor (ELPI), model ELPI+, Dekati] and Scanning Mobility Particle Spectrometry (SMPS, model 3080, TSI Inc.). A rich ethylene flame was chosen as it produces soot at a uniform rate. A co-flow diffusion flame burner was used, which comprised of two coaxial channels for the flow of fuel and air, respectively. Particle sampling is achieved with an N2-diluted sampling probe. The flow of dilution N2 through the annulus gap provides the negative pressure needed to sample the flue gas as well as cool down and dilute the particle-laden flow. The dilution ratio (DR) is controlled by N2 flow rate and the physical dimension of the annulus gap, and CO2 is chosen as the tracer gas for dilution ratio measurement. A high flow rate of dilution N2 (~45 SLPM) is applied and the dilution ratio is controlled to a range of 30 – 200. The probe samples the whole particle stream while maintaining quasi-isokinetic sampling, and is expected to cause little bias for particle measurements. The ELPI and SMPS (scanning mobility particle spectrometer) data show similar trends. The experiments were also conducted to know the efficiency of ELPI with respect to time to get the correct PSD (particle size distribution) data. It was found that after 2 minutes of measurements, ELPI have to be cleaned as it creates a virtual mode owing to the partly over filing of ELPI stages. In the future, a new dilution system will be used and PM data will be compared between SMPS and ELPI at dilution ratio of the order of 10000 using rich ethylene flames. At University of Maryland (UMD), the possible appearance of double luminous zones in normal diffusion flames was investigated in the literature and in the laboratory. Additional published flame lengths were added to our draft paper on flame lengths. Oxygen concentrations were measured in the chimney for inverse diffusion flames. A presentation on double luminous zones was made at ASGSR (American Society for Gravitational & Space Research) in Seattle. At UMD we tested IDFs (inverse diffusion flames) with CO2 as diluent in both fuel side and oxidizer side. These did not have double luminous zones. We also tested IDFs with Ar and He as diluent. These had the most remarkable double luminous zones we have seen to date. At UMD, work continues on improved imaging of species in normal and inverse diffusion flames. Interference filters have been acquired to allow the imaging of OH*, CH*, C2*, and CO2*. A Matlab script for Abel deconvolution has been modified for the resulting images. We have imaged our normal and inverse diffusion flames with various optical filters to identify the source of the blue emissions observed. We also have applied deconvolution to these images. Work has progressed on identifying the species responsible for double blue zones in normal and inverse gas jet diffusion flames. Bandpass filters allow imaging of C2* (515 nm), CO2* (455 nm), CH* (430 nm), and OH* (310 nm). The images are deconvolved, and corrections for CO2* emissions are made in the images of C2* and CH* emissions. WashU and UMD have been working together to begin welding B-type thermocouples (0.0015 in, 38 micron). These will initially be used to measure temperatures of co-flow diffusion flames at various stoichiometric mixture fraction. The injection port of the single droplet generator has been modified to allow for continuous operation with a Hencken flat flame. Droplet generation in the previous design was unstable when run with a flame. The single droplet generator will allow for the study of preferential vaporization in nonpremixed flames. We modified the design of the droplet generator to be able to maintain a steady droplet stream with flame. Drawings were sent to machine shop and parts have been modified. A new student has started at Wash U and he has been learning high-speed videography, PDPA, and manufacturing of fine-wire thermocouples. We are also developing a new method of making temperature measurements with high-speed videography. The spherical flame code has been demonstrated for the first time by the new team of Ph.D. students. Work at UMD has emphasized the numerical model. The current graduate students are now able to simulate spherical normal and inverse diffusion flames. They have implemented an improved method of choosing the initial temperature profile. The numerical modeling of spherical diffusion flames has progressed. This indicates that some normal flames of ethylene burning in air may be able to reach steady state. Furthermore, flames that continue burning with peak temperatures as low as 1000 K are observed. Previously no flames were predicted with these temperatures below 1100 K. We completed the final review of “Science Protocol on Intent to Conduct an International Collaborative Research Effort Entitled Flame Design,” to support collaborative research with Russian Scientists. Employing the principles developed by Long to support ACME (Advanced Combustion via Microgravity Experiments) research projects, WashU is developing capabilities to measure particle temperatures through direct photography.

	Based on recent findings, the new graduate student at Washington University is beginning to construct a tri-coflow burner to test a theory of applying Flame Design concepts to control heat flux. We have modified an existing burner to create a triaxial coflow flame. The inner nozzle supplies air or oxygen enriched air, the middle nozzle supplies fuel (methane) or nitrogen diluted fuel, and the outer nozzle supplies air or oxygen enriched air. The inner oxidizer creates an inner flame sheet, resulting in two distinct flame zones. The goal of this project is to maximize radiation heat transfer in methane combustion by using flame design concepts to create a soot favorable environment at the inner flame sheet and then oxidize soot particles before they are released to the ambient environment. We have been taking high speed videos of the flame sheet interactions and are preparing to take radiation measurements.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Abstracts for Journals and Proceedings	Wang Z, Sunderland PB, Axelbaum RL. "Double Blue Zones in Inverse Laminar Jet Diffusion Flames." Poster, 37th International Symposium on Combustion, Dublin, Ireland, July 29-August 3, 2018. 37th International Symposium on Combustion, Dublin, Ireland, July 29-August 3, 2018. , Jul-2018
Papers from Meeting Proceedings	Wang Z, Sunderland PB, Axelbaum RL. "Double Luminous Zones in Inverse Laminar Jet Diffusion Flames." 2018 Eastern States Section of the Combustion Institute Spring Technical Meeting, State College, PA, March 04-07, 2018. 2018 Eastern States Section of the Combustion Institute Spring Technical Meeting, State College, PA, March 04-07, 2018. , Mar-2018